

Does Sound Affect Typing Speed?

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Research Question:

Does the auditory environment significantly affect typing speed (WPM)?

Why this is Important:

- Most people prefer to listen to some sort of audio while working
- This study is beneficial to report productivity with typing speed
- With conclusions, we can understand a common optimization for a productive environment

The Design:

Design:

Complete Randomized Block Design (CRBD)

Fixed treatment factor (Audio)

Fixed blocking factor (The Tester)

- Testers have individual variation in their own typing speed range

Treatments:

- Lo-Fi
- Upbeat
- White Noise
- Silence as a control

Structure:

3 Participants x 4 Conditions x 3 Trials x 2 Replications = 72 total trials

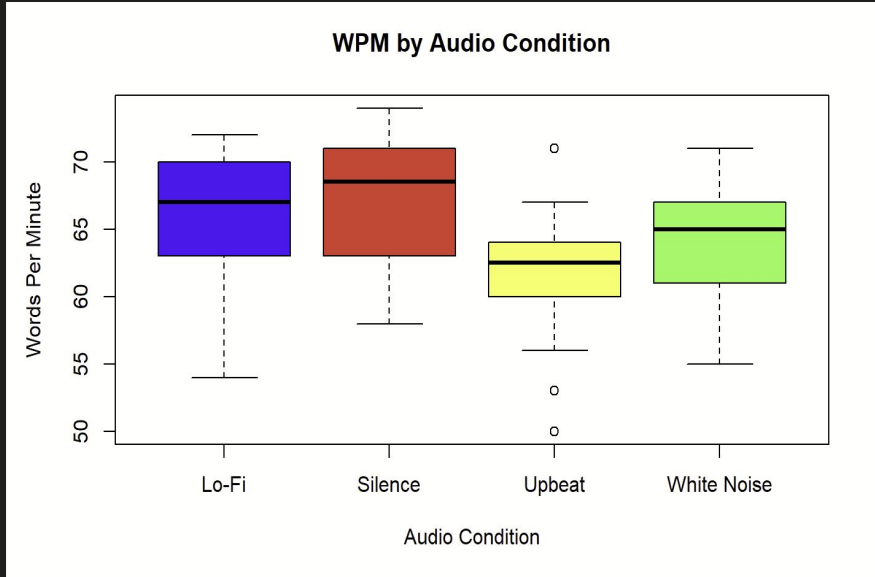
18 observations per audio treatment

Controls:

- [Typing.com](https://www.typing.com) for our one-minute tests
- Same headphones
- 60-65 dB volume setting
- Shuffled playlists to avoid bias
- Only one audio treatment a day (3 trials)

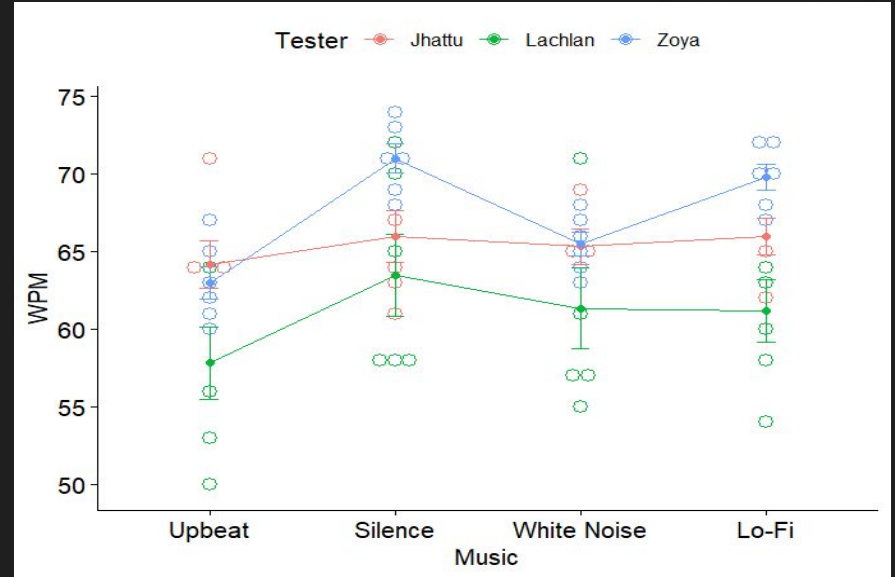
Intro Visualization

Box Plot



Silence is generally the best performing while upbeat is the worst.

Interaction

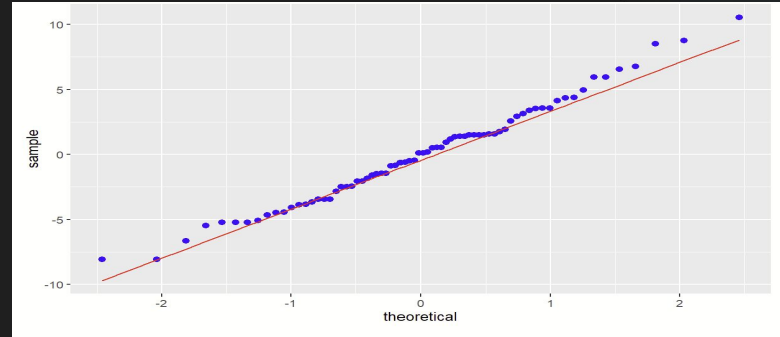


Each person has their individual typing range. The response to the conditions is similar (parallel).

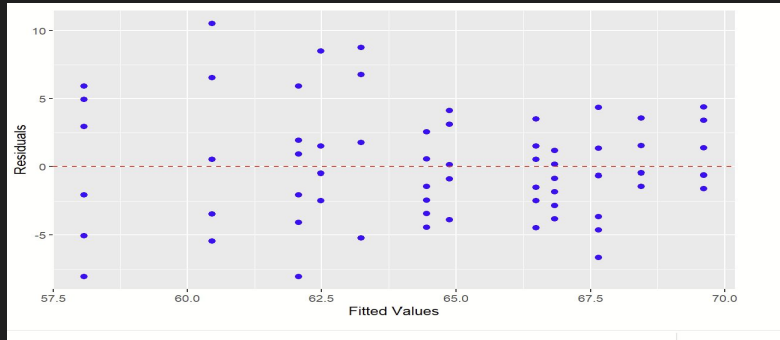
Model Conditions:

Independence: (each trial is an independent one minute test with the playlists shuffled so there is no bias)

Normality of Residuals: Q-Q Plot



Homogeneity of Variance: Residuals vs Fitted



Additivity: Interaction Plot

Running ANOVA:

Null Hypothesis: Auditory conditions have similar words per minute.

Alternative Hypothesis: At least one auditory condition has different words per minute.

Null Hypothesis: Testers have similar words per minute.

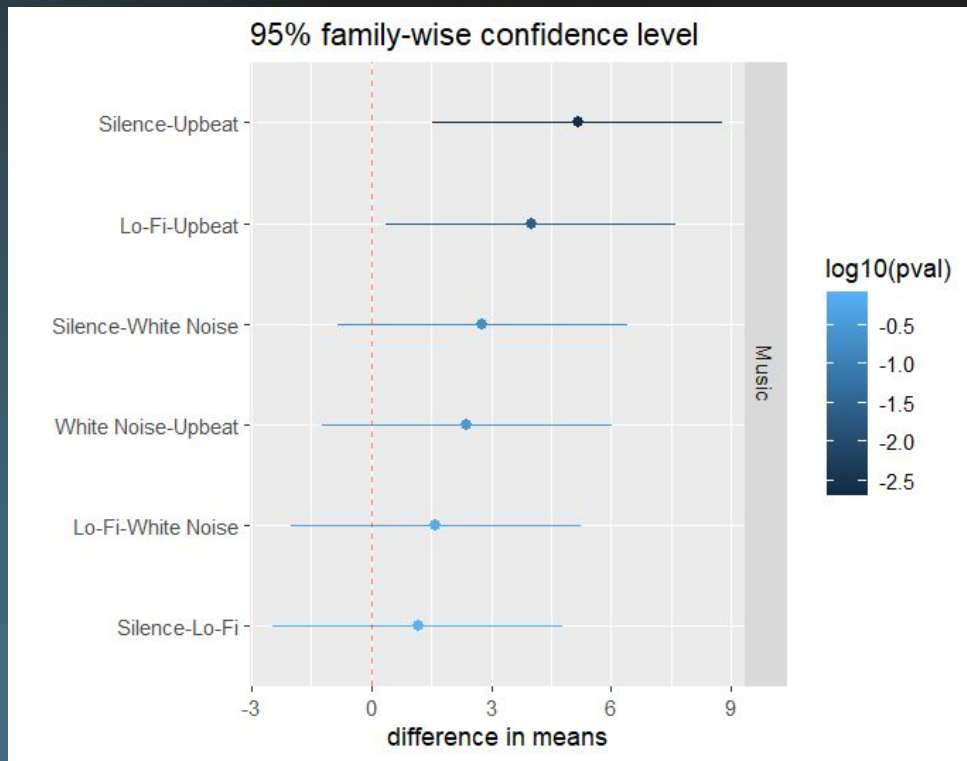
Alternative Hypothesis: At least one tester has different words per minute.

Both p-values are smaller than 0.05, rejecting both null hypotheses.

	DF	Sum Of Squares	Mean Squares	F value	P-value
Music	3	270.3	90.11	5.303	0.00246
Tester	2	511.9	255.93	15.06	0.00000409
Residuals	66	1121.6	16.99		

Tukey's HSD

- Upbeat significantly worse than Silence and Lo-Fi.
- Other comparisons result in insignificant difference.



Limitations

Short trial duration:

- One minute trials may not reflect sustained, real-world concentration

Individual Preference:

- The study assigned fixed audio conditions but did not account for participants' personal music tastes. Could influence how distracting a genre is.

Small Sample size:

- Only 3 participants (group members)

Limited Audio Categories:

- Only tested four specific fixed audio environments. Effects of other popular genres are unknown

Conclusion

Key Findings

- There is a significant effect on Words per Minute due to music.
- Upbeat music was the worst music to listen to while trying to type.
- Silence, white noise, and lo-fi were generally equivalent and optimal for typing.

Insights & Future Work

- We suspected the lyrics and fast rhythms in upbeat music disrupt the cognitive attention required for optimal typing.
- Results align with our initial prediction that upbeat music would be the most disruptive
- Test more participants across un-tested genres

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Thank You

Questions?